

TAL SCULLY

talscully@fas.harvard.edu ◇ 617-797-1875

EDUCATION

Harvard University, PhD Candidate in Systems, Synthetic, and Quantitative Biology (SSQB) 2018–2025

Advisor: Allon Klein, PhD, Professor of Systems Biology

Research: Using single-cell genomics to establish tunicate blood as a model for studying cell type evolution.

Massachusetts Institute of Technology, BS in Physics and in Theater Arts (GPA: 4.6/5.0) 2014–2018

Selected coursework:

Physics: Quantum Mechanics I II & III, Statistical Mechanics, Relativity, Junior Laboratory

Chemistry: Organic Chemistry I, Thermodynamics and Kinetics, Physical Chemistry II

Computer Science: Mathematics for Computer Science, Engineering Computation & Data Science

RESEARCH EXPERIENCE

Klein Lab, Harvard Dept. of Systems Biology, Graduate Student May 2019–Present

PI: Allon Klein, PhD, Professor of Systems Biology.

- Systematically testing hypotheses of cell type evolution using single-cell RNA sequencing (scRNA-seq) data of blood from eleven tunicate species.
- Modernized *Ciona robusta* blood cell classifications using scRNA-seq, live imaging, and in situ hybridization.
- Demonstrated dramatic evolutionary divergence between *Ciona robusta* and human immune cells.

Klein Lab, Harvard Dept. of Systems Biology, Undergraduate Researcher Jun–Aug 2017

PI: Allon Klein, PhD, Professor of Systems Biology.

Mentors: Caleb Weinreb, PhD and James Briggs, PhD.

Developed a computational method for quantifying the evolutionary conservation of gene expression patterns between frog and zebrafish development.

Field Group, MIT Dept. of Chemistry, Undergraduate Researcher Jun 2016–Mar 2017

PI: Robert Field, PhD, Haslam and Dewey Professor of Chemistry.

Mentor: Alex Hull, PhD.

Characterized synthesis of diatomic phosphorus using laser induced fluorescence and microwave spectroscopy.

Buchwald Group, MIT Dept. of Chemistry, Undergraduate Researcher Feb 2015–Jun 2016

PI: Stephen L. Buchwald, PhD, Camille Dreyfus Professor of Chemistry.

Mentor: Rana Kashif Khan, PhD.

Developed a synthesis of unnatural amino acids, and discovered an unknown intermediate reaction step.

PUBLICATIONS

Scully TD, Pickett CJ, Gort-Freitas NA, Davidson B, Klein AM. Divergence of immune cell types in chordate blood. *Current Biology*. 2025 Dec 15;35(24):5938-5954.e12. Available from: <http://dx.doi.org/10.1016/j.cub.2025.10.032>

Scully TD, Klein AM. A mannitol-based buffer improves single-cell RNA sequencing of high-salt marine cells. *BMC Genomics*. 2025 Dec 1;27(1):9. Available from: <http://dx.doi.org/10.1186/s12864-025-12370-7>

Lebel M, Sancerini T, Rabiteau S, Marchal S, Sharma P, **Scully TD**, et al. Multi-stage single-cell atlas of *Botryllus schlosseri* asexual development unveils dedifferentiating bud founder cells. *bioRxiv*. 2025. p. 2025.05.27.655565. Available from: <https://www.biorxiv.org/content/10.1101/2025.05.27.655565v1>

Gruner HN, Pickett CJ, Bao JY, Garcia R, Hozumi A, **Scully TD**, et al. Neural signaling contributes to heart formation and growth in the invertebrate chordate, *Ciona robusta*. *bioRxiv*. 2025. p. 2025.04.28.651085. Available from: <https://www.biorxiv.org/content/10.1101/2025.04.28.651085v1>

Khan RKM, Zhao Y, **Scully TD**, Buchwald SL. 2018. Catalytic Arylhydroxylation of Dehydroalanine in Continuous Flow for Simple Access to Unnatural Amino Acids. *Chemistry* 24 (57): 15215-18.

PRESENTATIONS

International Tunicate Meeting, Santa Cruz, CA Jul 2024
Invited Talk: "Evolution of cell types in tunicate blood"

ASCB Cell Bio 2023, Boston, MA Dec 2023
Poster: "Evolution of Innate Immune Cells in the Urochordate *Ciona robusta*."

Harvard Systems Biology Pizza Talk, Boston, MA Oct 2023
Talk: "Evolution of Innate Immune Cells in the Urochordate *Ciona robusta*."

EMBL The Identity and Evolution of Cell Types, Virtual May 2021
Poster: "The evolution of vertebrate red blood cells."

AWARDS AND RECOGNITION

International Tunicate Meeting, *Invited Talk* Jul 2024

Harvard Systems Biology Dept., *Lynch Fellow* 2022

Harvard Certificate of Distinction in Teaching, *for the Fall 2019 Semester* Apr 2020

NSF Graduate Research Fellowship Program, *Honorable Mention* Mar 2018

LEADERSHIP AND SERVICE

Development, *Ad Hoc Reviewer* Mar 2024–Present

New York Times, *Published Letter to the Editor* May 2025
Published a short letter objecting to the Trump administration's attacks on Harvard funding and international students. Available at <https://www.nytimes.com/2025/05/29/opinion/harvard-trump.html>

Harvard Systems Biology Dept., *Diversity Equity and Inclusion Fellow* Jan–Jun 2023
Established a Peer Mentorship Program to support first-year students in the SSQB PhD program.

Scientific Citizenship Initiative, *Leadership Team* Mar 2022–Jun 2024
Advised the leadership of the Harvard Scientific Citizenship Initiative on current and future directions.

Social Issues in Biology "Race in Biosciences" Curriculum, *Co-Author* Sep 2020–Jan 2023
Created a journal club series covering a range of topics at the intersection of race, biology, and academia. Available at <https://projects.iq.harvard.edu/race-in-biosciences>

Social Issues in Biology Journal Club, *Co-Lead* Sep 2019–May 2023
Organized journal clubs for the Harvard community on subjects at the intersection of science and society.

SSQB Equity and Inclusion Working Group, *Co-Founder and Chair* Jun 2020–Dec 2022
Co-wrote the SSQB Diversity Statement and advocated for structures to promote equity and belonging.

SSQB Diversity in Graduate Admissions Working Group, *Co-Founder and Chair* Jul 2020–Sep 2021
Established an Application Assistance Program for applicants from underrepresented backgrounds in STEM.

MENTORSHIP AND TEACHING

Student Mentoring

Cohen Manges, Undergraduate Researcher based at Swarthmore College (remote)	<i>Oct–Dec 2024</i>
Natalia Orlovsky, Graduate Rotation Student	<i>Apr–May 2023</i>
Aashna Shah, Graduate Rotation Student	<i>Feb–Mar 2021</i>
Hai Pham, Undergraduate Researcher, with Co-Mentor Laura Bagamery, PhD	<i>Jun–Aug 2021</i>

Mentoring Triads, Harvard Genetics Postdoc Group, *Invited Speaker* *Mar 2024*
Created and ran a workshop on approaching difficult conversations in mentoring relationships.

Communication of Science Class, Harvard Dept. of Systems Biology, *Teaching Fellow* *Sep–Dec 2019*
Taught graduate students the fundamentals of science writing, figure design, and oral presentation.

Junior Laboratory Class, MIT Dept. of Physics, *Teaching Assistant* *Sep–Dec 2017*
Taught undergraduates fundamental experiments in modern physics, along with error analysis and writing.

OUTREACH

Civic Science Clinic, *Community Organization Fellow* *May–Aug 2023*
Gained hands-on community engagement experience with the nonprofit Essential Partners.

“What is Systems Biology,” Cambridge Science Festival, *Co-Lead* *Apr 2019*
Created an event for the Cambridge Science Festival with interactive activities for K–12 students.

MIT Museum Girl’s Day, *Invited Speaker* *Nov 2018*
Spoke to kids aged 10+ about using optical tweezers to study mechanical properties of biological molecules.

SPLASH, MIT Educational Studies Program, *Volunteer Teacher* *Annually, Nov 2014–2018*
Developed and taught classes to high school students on computational biology, crystal field theory, statistical mechanics, special relativity, and improv comedy.